

Python Beginner to Advanced – Full Resources Roadmap 🚀

1. Python Start ಮಾಡಲು (Beginner Level)

Install Python

- [Python Official Website](#)
- [VS Code](#)

Free Beginner Courses

1. [Python for Beginners – freeCodeCamp YouTube](#)
 2. [Programming with Mosh – Python Tutorial](#)
 3. [W3Schools Python Tutorial](#)
 4. [GeeksforGeeks Python](#)
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2. Basics You Must Learn

- ✓ Variables
- ✓ Data Types
- ✓ Input / Output
- ✓ Conditions (`if else`)
- ✓ Loops (`for, while`)
- ✓ Functions
- ✓ Lists / Tuples / Dictionaries / Sets
- ✓ Strings
- ✓ File Handling
- ✓ Exception Handling
- ✓ OOPs (Class & Objects)

Practice Websites

- [HackerRank Python](#)
- [LeetCode](#)
- [CodeChef](#)

3. Intermediate Python

Learn These Topics

- Modules & Packages
- Lambda Functions
- List Comprehension
- Decorators
- Generators
- Iterators
- Virtual Environment
- APIs
- JSON
- Multithreading

Best Resources

- [Real Python](#)
 - [Python Docs](#)
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4. Advanced Python

Topics

- 🔥 Async Programming
- 🔥 Memory Management
- 🔥 DSA in Python
- 🔥 Design Patterns
- 🔥 Testing ([pytest](#))
- 🔥 Automation
- 🔥 Web Scraping
- 🔥 Networking
- 🔥 System Design Basics

Advanced Resources

- [Advanced Python Playlist](#)
 - [Automate the Boring Stuff with Python](#)
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5. Python + AI/ML Roadmap

Since you are interested in AI/Embedded projects:

Learn Libraries

Data Science

- NumPy
- Pandas
- Matplotlib

AI/ML

- Scikit-learn
- TensorFlow
- PyTorch

Resources

- [Kaggle Learn](#)
 - [DeepLearning.AI](#)
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6. Python + Embedded Systems

Learn

- ✓ Raspberry Pi
- ✓ ESP32 + MicroPython
- ✓ OpenCV
- ✓ IoT Projects
- ✓ Camera Streaming
- ✓ Sensor Integration

Resources

- [Raspberry Pi Documentation](#)
 - [OpenCV Python Tutorials](#)
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7. Best YouTube Channels

- [freeCodeCamp](#)
 - [CodeWithHarry](#)
 - [Programming with Mosh](#)
 - [Tech With Tim](#)
 - [Corey Schafer](#)
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8. Best Python Books

1. Python Crash Course
 2. Automate the Boring Stuff with Python
 3. Fluent Python
 4. Effective Python
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9. Best Real Projects for Resume

Beginner

- Calculator
- To-Do App
- Student Management System

Intermediate

- Face Detection
- Weather App
- Chat Application

- File Encryptor

Advanced

- AI Voice Assistant
 - Smart Surveillance System
 - Drone Monitoring
 - Raspberry Pi Live Streaming
 - Defense Dog Camera System
 - AI Disease Detection App
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10. Final Learning Order

Python Basics



OOP + File Handling



DSA + Problem Solving



Intermediate Python



Projects



Git & GitHub



AI / ML or Embedded



Advanced Projects

GitHub Learning

- [GitHub Docs](#)
 - [Git Handbook](#)
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Daily Routine (Best)

- 🕒 1 Hour → Python Theory
 - 🕒 2 Hours → Coding Practice
 - 🕒 1 Hour → Projects
 - 🕒 30 Min → GitHub Upload
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